The documentation and process conversion measures necessary to comply with this revision shall be completed by 18 November 2000.

INCH-POUND

MIL-PRF-19500/313E
18 August 2000
SUPERSEDING
MIL-PRF-19500/313D
21 August 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW POWER
TYPES 2N2432, 2N2432A, JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for low power, high speed chopper, NPN silicon transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for die.

1.2 Physical dimensions. See figure 1 (T0-18) and figures 2 and 3 (JANHC/JANKC die).

1.3 Maximum ratings.

Type	P_T T_A = +25°C	V_{CBO}	V_{CEO}	V_{ECO}	I_C	T_{STG}	T_J	$R_{\theta JA}$
	mW	V dc	V dc	V dc	mA dc	°C	°C	°C/mW
2N2432	500 (1)	30	30	15	100	-65 to +200	-65 to +175	325
2N2432A	500 (1)	45	45	18	100	-65 to +200	-65 to +175	325

(1) Derate linearly 3.08 mW/°C above T_A > +37.5°C.

1.4 Primary electrical characteristics at T_A = 25°C.

Limits	h_{FE1} V_{CE} = 5 V dc I_C = 10 µA dc	h_{FE2} V_{CE} = 5 V dc I_C = 1 mA dc	$h_{FE(inv)1}$ V_{CE} = 10 mA dc I_E = 200 µA dc		$V_{CE(sat)}$ I_C = 10 mA dc I_B = 500 µA dc	$r_{ec(on)}$ I_e = 100 µA ac (rms) I_B = 1 mA dc, I_E = 0, f = 1 kHz		$R_{\theta JC}$
			2N2432	2N2432A		2N2432	2N2432A	
					V dc	Ohms	Ohms	°C/mW
Min	30	80	2	3				
Max		400			.15	20	15	.25

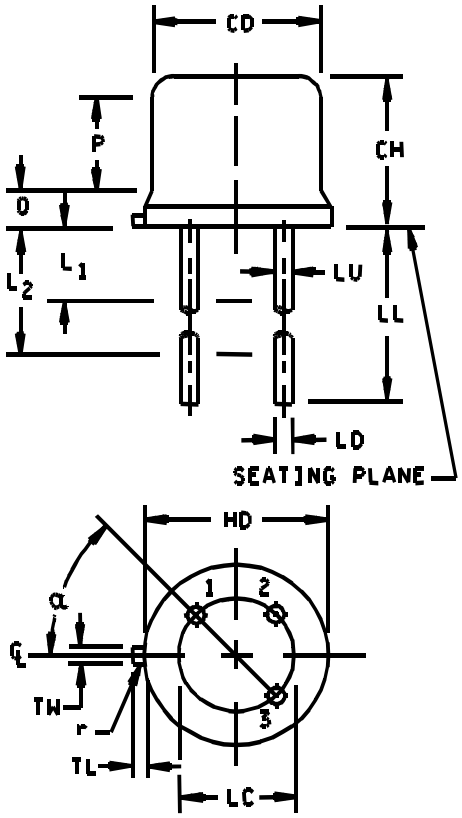
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC/VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8
LU	.016	.019	0.41	0.48	7,8
L1	---	.050	---	1.27	7,8
L2	.250	---	6.35	---	7,8
P	.100	---	2.54	---	
Q	---	.030	---	0.76	5
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r	---	.010	---	0.25	10
α	45° TP		45° TP		6

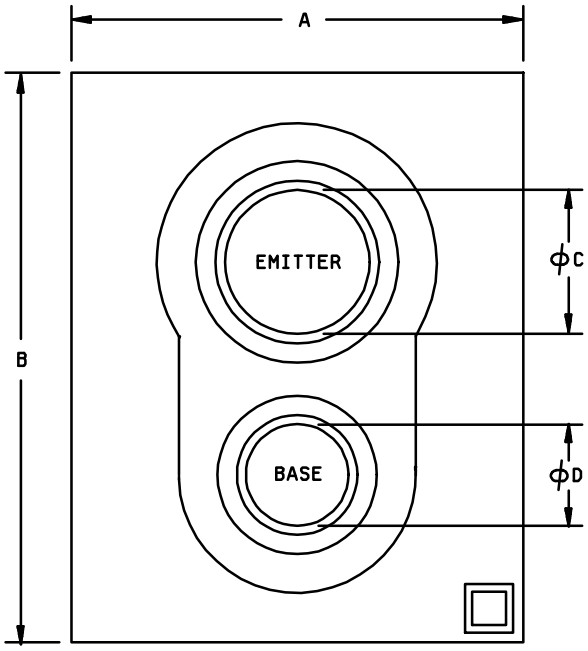


NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions (similar to TO-18).

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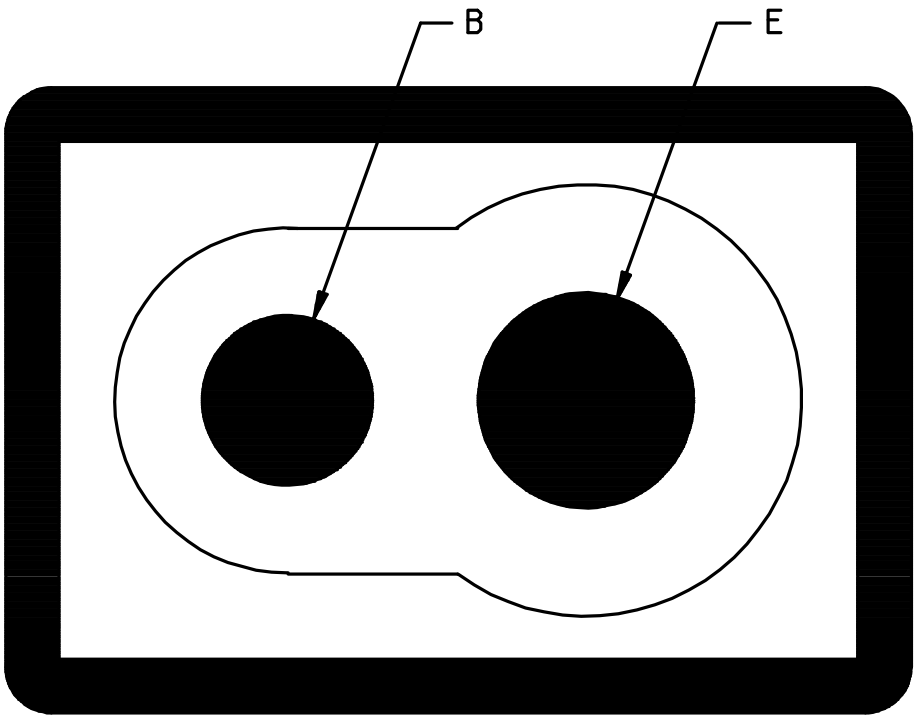
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.013	.017	.330	.430
B	.017	.021	.430	.533
ϕC		.0048		.122
ϕD		.0034		.086



NOTES:
1. Chip thickness: 5 to 7 mils
2. Top metal: Aluminum 10,000 Å min, 12,000 Å nominal
3. Back metal: Gold 2,000 Å min, 4,000 Å nominal
4. Backside: Collector

FIGURE 2. JANHC (A-version) die dimensions.

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NOTES:

1.	Chip size	15 x 19 mils ± 1 mil
2.	Chip thickness	10 ± 1.5 mil
3.	Top metal	Aluminum 15,000Å minimum, 18,000Å nominal
4.	Back metal	A. Gold 2,500Å minimum, 3,000Å nominal B. Eutectic Mount – No Gold
5.	Backside	Collector
6.	Bonding pad	B = 3 mils, E = 4 mils diameter
7.	Passivation	Si_3N_4 (Silicon Nitride) 2 kÅ min, 2.2 kÅ nom.

FIGURE 3. Physical dimensions, JANHCA and JANKCA (B - version) die.

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1.4 Primary electrical characteristics at T_A = 25°C - continued

Limits	$V_{CE(ofs)1}$ I_B = 200 µA dc I_E = 0		$V_{EC(ofs)2}$ I_B = 1 mA dc I_E = 0		*h_{fe}* V_{CE} = 5 V dc I_C = 1 mA dc f = 20 MHz	C_{obo} V_{CB} = 0 mA dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	C_{ibo} V_{EB} = 0 I_C = 0 100 kHz ≤ f ≤ 1 MHz
	2N2432	2N2432A	2N2432	2N2432A			
	mV dc	mV dc	mV dc	mV dc		pF	pF
Min					2		
Max	0.5	0.4	1.0	0.7	10	12	12

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.4).

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3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$V_{(BR)ECO}$ - - - Breakdown voltage, emitter to collector, with base open circuited.

$h_{FE(inv)}$ - - - Forward current transfer ratio except that the collector and emitter shall be interchanged.

I_e - - - - - Emitter current (rms)

$r_{ec(on)}$ - - - Small signal emitter to collector on state resistance.

V_{BC} - - - - Base to collector voltage.

$V_{EC(ofs)}$ - - - Emitter to collector offset voltage, i.e., open circuit voltage between emitter and collector when the base to collector junction is forward biased.

V_{ec} - - - - Emitter to collector voltage (rms).

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1, 2, and 3.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking may be omitted from the body, but shall be retained on the initial container.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table 1.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC devices. JANHC devices shall be qualified in accordance with MIL-PRF-19500.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{CBO2}, h_{FE1}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO2}; h_{FE1}; ΔI_{CBO2} = 100% of initial value or 5 nA dc, whichever is greater. Δh_{FE1} = ±15%	I_{CBO2} , h_{FE1}
12	See 4.3.1 240 hours minimum	See 4.3.1 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15%	Subgroup 2 of table I herein; ΔI_{CBO2} = 100% of initial value or5 nA dc, whichever is greater; Δh_{FE1} = ±15%

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements in 4.5 of MIL-STD-750:

V_{CB} = 10-30 V dc; Power shall be applied to achieve T_J = 135°C minimum and a minimum power dissipation (P_D) ≥ 75 % of maximum rated P_T (see 1.3).

4.3.2 Screening JANHC. Screening shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100 percent probed to ensure compliance with group A, subgroup 2.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131 of MIL-STD-750.

a. I_M measurement current ----------1 mA.

b. I_H forward heating current ----------30 mA (min).

c. t_H heating time ----------25 - 30 ms.

d. t_{md} measurement delay time ---------60 µs max.

e. V_{CE} collector-emitter voltage ---------5 V dc minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 250°C/W.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and table III herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with group A, subgroup 2 and table III herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc
B5	1027	V_{CB} = 10 V dc; T_A = +125°C ± 25°C for 96 hours adjusted as required according to the chosen T_A to give a T_J = +275°C minimum. Optionally, the test may be conducted for minimum 216 hours with P_T adjusted to achieve T_J = 225°C minimum, sample size (for option) = 45, c = 0. In this case, the ambient temperature shall be adjusted such that a minimum 75 percent of maximum rated P_T (see 1.3) is applied to the device under test. (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.)

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 -30 V dc, Power shall be applied to achieve T_J = 150°C minimum and a minimum power dissipation $P_D \geq 75\%$ of maximum rated P_T as defined in 1.3 herein.
2	1039	The steady state life test of step 1 shall be extended to 1,000 hrs for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B, step 2 shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0

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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS).and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and table III herein.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	See 4.5.7
C6	1026	1,000 hours at V_{CB} = 10 to 30 V dc; power shall be applied to achieve T_J = 150°C minimum and a minimum power dissipation $P_D \geq 75\%$ of maximum rated P_T as defined in 1.3 herein.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	See 4.5.7
C6		Not applicable

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E Inspection. Group E inspection shall be performed for qualification or re-qualification only. The tests specified in table II herein must be performed to maintain qualification.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Emitter to collector cutoff current. Method of test shall be in accordance with method 3041 of MIL-STD-750, condition C, except that all references to the collector and emitter of the transistor under test shall be interchanged.

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4.5.3 Emitter to collector breakdown voltage. Method of test shall be in accordance with method 3011 of MIL-STD-750, condition D, except that all references to the collector and emitter of the transistor under test shall be interchanged.

4.5.4 Forward current transfer ratio (inverted connection). Method of test shall be in accordance with method 3076 of MIL-STD-750, except that all references to the collector and emitter of the transistor under test shall be interchanged. Then $h_{FE(inv)} = I_E / I_B$.

4.5.5 Emitter to collector offset voltage. The transistor shall be tested in the circuit on figure 4. The base current shall be adjusted to the specified value. The voltage between the emitter and collector shall be measured using a voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement.

4.5.6 Small signal emitter to collector on-state resistance. The transistor shall be tested in the circuit of figure 5. The base current shall be adjusted to the specified value and an ac sinusoidal signal current, I_e , of the specified rms value shall be applied between the emitter and collector. The rms voltage V_{ec}, between the emitter and collector shall be measured using an ac voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement. The small signal emitter to collector on-state resistance shall then be determined as follows:

$$r_{ec(on)} = V_{ec} / I_E$$

Where V_{ec} is the rms voltage between the emitter and collector.

4.5.7 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 30 mA dc.

b. Collector emitter voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be selected with $25°C \leq T_R \leq 75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit for $R_{\theta JC}$ shall be 0.25°C/mW.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agencys' automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071	n = 45 devices, c = 0				
Solderability 3/, 5/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/, 4/, 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/, 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 5/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 5/		Group A, subgroup 2				
Bond strength 3/, 5/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = 300°C at t = 2 hrs n = 11 wires, c = 0				
Subgroup 2						
Breakdown voltage collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N2432				30		V dc
2N2432A				45		V dc
Collector to base cutoff current	3036	Bias condition D;	I_{CBO1}			
2N2432		V_{CB} = 30 V dc			100	µA dc
2N2432A		V_{CB} = 45 V dc			100	µA dc
Breakdown voltage emitter to collector	3011	Bias condition D; I_E = 100 µA dc; I_E = 0 (see 4.5.3)	$V_{(BR)ECO1}$			
2N2432				15		V dc
2N2432A				18		V dc
Breakdown voltage emitter to collector	3011	Bias condition D; I_E = 10 mA dc; I_B = 0; pulsed (see 4.5.1and 4.5.3)	$V_{(BR)EC02}$	10		V dc
Collector to base cutoff current	3036	Bias condition D;	I_{CBO2}			
2N2432		V_{CB} = 25 V dc			10	nA dc
2N2432A		V_{CB} = 40 V dc			10	nA dc
Collector to emitter cutoff current	3041	Bias condition C;	I_{CES1}			
2N2432		V_{CB} = 25 V dc			10	nA dc
2N2432A		V_{CB} = 40 V dc			10	nA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - continued						
Emitter to collector cutoff current	3041	Bias condition C; V_{EC} = 15 V dc; V_{BC} = 0; (see 4.5.2)	I_{ECS1}		2	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 15 V dc	I_{EBO}		2	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc;	h_{FE1}	30		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 mA dc	h_{FE2}	80	400	
Forward-current transfer ratio (inverted connection)	3076	V_{CE} = 5 V dc; I_C = 0.2 mA dc;(see 4.5.4)	$h_{FE(inv)1}$			
2N2432 2N2432A				2 3		
Saturation voltage and resistance	3071	I_C = 10 V dc; I_B = 0.5 mA dc;	$V_{CE(sat)}$		0.15	V dc
Emitter to collector offset voltage		I_B = 200 µA dc; I_E = 0 mA dc (see 4.5.5 and figure 5)	$V_{EC(ofs)1}$			
2N2432 2N2432A					0.5 0.4	mV dc mV dc
Emitter to collector offset voltage		I_B = 1 mA dc; I_E = 0 mA dc (see 4.5.5 and figure 5)	$V_{EC(ofs)2}$			
2N2432 2N2432A					1.0 0.7	mV dc mV dc
Subgroup 3						
High-temperature operation:		T_A = +125°C				
Collector to emitter cutoff current	3041	Bias condition C;	I_{CES2}			
2N2432 2N2432A		V_{CB} = 25 V dc V_{CB} = 40 V dc			250 250	nA dc nA dc
Emitter to collector cutoff current	3041	Bias condition C; V_{EC} = 15 V dc; V_{BC} = 0 (see 4.5.2)	I_{ECS2}		200	nA dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc; pulsed (see 4.5.1)	h_{FE3}	10		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 mA dc; pulsed (see 4.5.1)	h_{FE4}	25		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - continued						
Forward-current transfer ratio (inverted connection)	3076	V_{CE} = 5 V dc; I_E = 200 µA dc; pulsed (see 4.5.1 and 4.5.4)	$h_{FE(inv)2}$	1.8		
Subgroup 4						
Small signal emitter collector on state resistance		I_B = 1 mA dc; I_E = 0; I_e = 100 µA ac (rms); f = 1 kHz (see 4.5.6 and figure 5)	$r_{ec(on)}$			
2N2432					20	Ω
2N2432A					15	Ω
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 20 MHz	$\|h_{fe}\|$	2	10	
Open circuit output capacitance	3236	V_{CB} = 0; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		12.0	pF
Input capacitance	3240	V_{EB} = 0 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		12.0	pF
Subgroups 5 and 6						
Not applicable						
Subgroup 7 5/						
Decap internal visual (design verification)	2075	n = 1 device, c = 0				

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ Not required for JANS devices.

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TABLE II. Group E inspection (all quality levels) - For qualification only

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See group A, subgroup 2 and table III herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles, $\Delta T_J \geq +100°C$; forced air cooling allowed on cooling cycle only.	
Electrical measurements		See group A, subgroup 2 and table III herein.	
Subgroup 3			
Not applicable			
Subgroup 4			
Not applicable			
Subgroup 5			
Not applicable			

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TABLE III. Groups B and C delta measurements.

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Collector-base cutoff current 2N2432 2N2432A	3036	Bias condition D V_{CB} = 25 V dc V_{CB} = 40 V dc	ΔI_{CB02} 1/	100% of initial value or 5 nA dc, whichever is greater.	
2	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc; pulsed see 4.5.1	Δh_{FE2} 1/	±25% change from initial reading.	

1/ Devices which exceed the group A limits for this test shall not be accepted.
2/ The delta measurements for group B, table VIa (JANS) of MIL-PRF-19500 are as follows:
a. Subgroup 5, see table III herein, steps 1 and 2.
b. Subgroup 3, see table III herein, steps 1, 2, 3, 4, and 5.
c. Subgroup 6, see table III herein, steps 1, 2, 3, 4, and 5.
3/ The delta measurements for 4.4.2.2 herein (group B, JAN, JANTX, and JANTXV) are as follows: Steps 1 and 2 of table III shall be performed after each step in 4.4.2.2 herein.
4/ The delta measurements for group C, table VII of MIL-PRF-19500 are as follows: Subgroup 6, see table III herein, steps 1 and 2 for JANS only.

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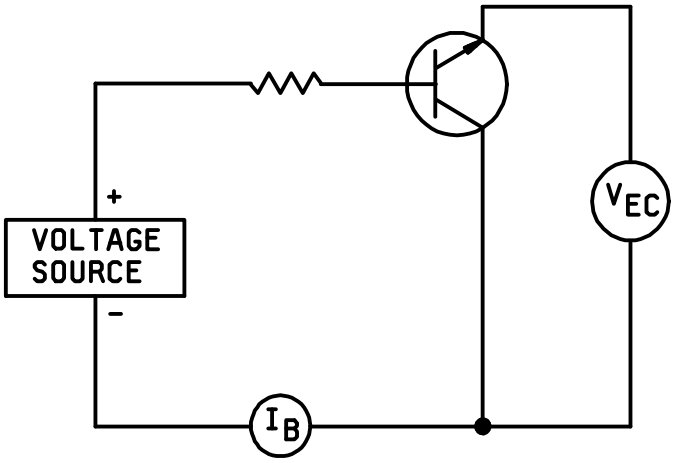


FIGURE 4. Emitter to collector offset voltage test circuit.

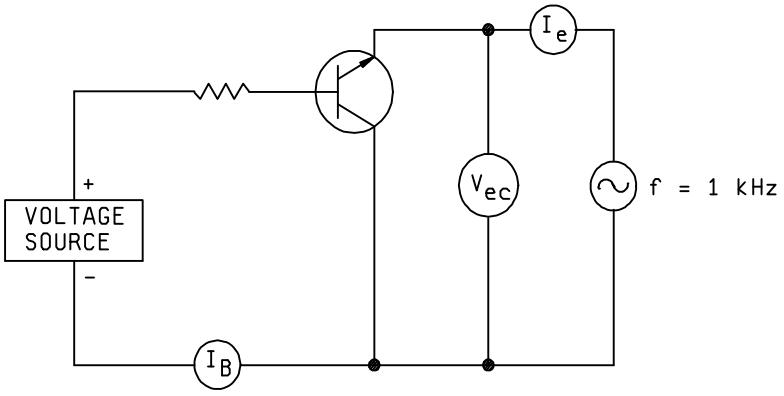


FIGURE 5. Small-signal emitter-collector on-state resistance test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2).

c. Lead finish (see 3.4.1).

d. Type designation and product assurance level.

e. Packaging requirements (see 5.1).

6.3 Suppliers of JANHC die. The qualified JANC suppliers with the applicable letter version (example, JANHCA2N2432 or JANHCA2N2432A) will be identified on the QML.

JANC ordering information		
PIN	Manufacturer	
	12498	43611
2N2432 2N2432A	JANHCA2N2432 JANHCA2N2432A	JANHCB2N2432 JANHCB2N2432A
2N2432 2N2432A		JANKCB2N2432 JANKCB2N2432A

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - MI
Navy - AS, CG, MC
Air Force - 19

Preparing activity:
DLA - CC

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/313E	2. DOCUMENT DATE 18 August 2000

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW POWER TYPES 2N2432, 2N2432A, JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center, Columbus ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888			

DD Form 1426, Feb 1999 (EG) Previous editions are obsolete WHS/DIOR, Feb 99

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